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GMFRMNet: A Global-Local Multiscale Feature Refinement Mamba Network for Cropland Change Detection

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- A Global-Local Multiscale Feature Refinement Mamba Network (GMFRMNet) is designed for satellite-based cropland change detection.
- A Global-Local Context State-Space Block (GLCSB) is designed to improve true change discrimination under complex temporal conditions.
- A Multiscale Channel-Spatial Refinement Block (MCSRB) is designed to enhance small, irregular and dispersed cropland change extraction.

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GMFRMNet: A Global-Local Multiscale Feature Refinement Mamba Network for Cropland Change Detection

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ABSTRACT

Cropland change detection (CD) is of great significance for cropland protection, land management and agricultural sustainability. However, cropland CD is susceptible to phenological variations, illumination changes and imaging-condition differences, which make the feature relationships between changed and unchanged regions more complex and limit the contextual discrimination capability of existing CD methods for actual cropland change regions. Meanwhile, cropland changes usually exhibit large variations in scale, irregular shapes and dispersed spatial distributions, which easily lead to missed detections and incomplete extraction of local change regions. To address these issues, this paper proposes a Global-Local Multiscale Feature Refinement Mamba Network (GMFRMNet) for cropland CD. GMFRMNet adopts a weight-shared Siamese Visual Mamba (VMamba) encoder to extract bi-temporal remote sensing image features. Specifically, a Global-Local Context State-Space Block (GLCSB) is designed to capture long-range spatial dependencies of bi-temporal interaction features through global relation modeling, while enhancing local texture and spatial information through a local branch, thereby improving the contextual discrimination capability for actual cropland change regions. In addition, a Multiscale Channel-Spatial Refinement Block (MCSRB) is designed to extract contextual features with different receptive fields through multiscale convolutions and to integrate channel feature enhancement with spatial position refinement to improve the representation of small and irregular cropland change regions, thereby alleviating missed detections and local incompleteness of changed regions. Experimental results on the CLCD, JL-1 and PX-CLCD cropland CD datasets show that GMFRMNet outperforms eight representative CD methods, achieving F1 scores of 81.65%, 82.23% and 96.60%, respectively.

1. Introduction

Cropland is an essential land resource for ensuring food production, maintaining ecological balance and supporting sustainable land development (Potapov et al., 2022). In recent years, human activities associated with rapid urbanization, industrial expansion and ecological construction have intensified. Activities such as non-agricultural construction, afforestation and artificial lake excavation have occupied and damaged large areas of cropland, posing serious threats to food security and agricultural sustainability (Venter et al., 2026). Therefore, accurate and timely cropland change detection (CD) is of great significance for cropland protection, land resource management and agricultural sustainability.

CD refers to a technique that extracts land cover change information by comparing and analyzing remote sensing images acquired over the same area at different times. Early CD methods mainly included image algebra-based methods (Gong et al., 2012; Lu et al., 2004; Radke et al., 2005), image transformation-based methods (Celik and Ma, 2010; Kuncheva and Faithfull, 2014; Ridd and Liu, 1998) and machine learning-based methods (Feng et al., 2018; Nemmour and Chibani, 2006; Saha et al., 2019). Although these methods can achieve certain accuracy in CD tasks, they rely heavily on empirical thresholds and handcrafted feature design, resulting in limited robustness and generalization ability. In addition, they are limited in modeling complex spatial contextual information. With the development of deep learning (Ding et al., 2025; Lv et al., 2022; Zhu et al., 2017), Convolutional Neural Networks (CNNs) have been widely applied to CD tasks by extracting multiscale semantic information from images through convolution operations at different scales (Daudt et al., 2018; Gong et al., 2017; Hou et al., 2017). Based on CNN architectures, strategies such as attention mechanisms (Peng et al., 2021; Song et al., 2023), multiscale convolutions (Chen et al., 2019; Guo et al., 2022), pyramid pooling (Li et al., 2024; Yang et al., 2022) and dense scale connections (Fang et al., 2022; Miao et al., 2024) are further introduced to enhance feature extraction and improve CD accuracy. However, due to the local receptive fields of convolution operations, CNNs have limited ability to model long-range dependencies and global spatial context in high-resolution images, which still restricts their performance in complex CD scenarios. To compensate for the insufficient global context modeling capability of CNNs, transformers have been introduced into CD tasks (Bandara and Patel, 2022; Chen et al., 2022; Zhang et al., 2022). Transformers model long-range dependencies and global spatial relationships through the self-attention mechanism, thereby improving the feature extraction capability for change regions in complex scenes. However, transformer-based methods still have certain limitations in CD tasks. On the one hand, the computational complexity of the self-attention mechanism increases quadratically with the input sequence length, which easily leads to high computational and memory costs

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in CD tasks. On the other hand, transformers lack the inherent local receptive fields of CNNs and show insufficient capability in multiscale feature extraction (Li et al., 2022; Liu et al., 2022; Zhu et al., 2025).

Mamba achieves global context modeling with linear complexity through the selective scanning mechanism and maintains a large global receptive field while reducing computational costs, thereby providing a new architectural solution to alleviate the limited local receptive field of CNNs and the high computational complexity of transformers (Chen et al., 2024b; Xiao et al., 2025; Zhang et al., 2026). Therefore, Mamba has been increasingly applied to CD tasks. Chen et al. (2024a) proposed ChangeMamba, the first attempt to introduce the Mamba architecture into CD. It enhances the interaction between bi-temporal features through spatiotemporal relationship modeling and demonstrates the potential of Mamba in CD tasks. Zhang et al. (2025) developed CDMamba, in which a scaled residual convolutional Mamba module and a global-local guided fusion module are designed to enhance local information and improve the extraction of bi-temporal change features. Liu et al. (2026) proposed a global-local Mamba network to address the difficulty of sufficiently extracting local change information from global features under complex conditions. Through joint global-local modeling, combined with multiscale semantic aggregation, bi-temporal fusion attention and a cross-spatial interaction enhancement module, the network improves the extraction capability for change regions in complex scenes. Wang et al. (2025) proposed a Siamese pyramid Mamba network to address the problems of complex textures, multiscale spatial structures and severe imbalance between changed and unchanged pixels in high-resolution images. In this network, a hybrid CNN and Mamba backbone is designed to capture both global and local information from bi-temporal features, thereby improving CD accuracy. Sun et al. (2025) introduced a difference enhancement and cross-scale interactive fusion Mamba network to address the insufficient ability of Mamba to preserve shallow texture and local detail information. By using a difference feature enhancement module and a cross-scale interactive fusion strategy, this network improves the representation of local contextual details and enhances the fusion capability for change features across different scales. Wu et al. (2025) proposed a Mamba-driven domain adaptation network to address pseudo changes, false alarms and missed detections caused by domain shifts between bi-temporal images in cropland CD, which are induced by factors such as phenological variation, sensor differences, illumination changes and atmospheric interference. Through a cross-domain alignment mechanism, the network suppresses pseudo changes and enhances the discrimination capability for true change regions. In summary, existing Mamba-based CD methods have shown certain advantages in modeling long-range spatial dependencies and representing global context. However, there is still room for improvement in local detail preservation, bi-temporal feature interaction and multiscale change feature representation. In cropland CD, bi-temporal images are easily affected by phenological variation, illumination differences and differences in imaging conditions. These factors make the feature relationships between changed regions and surrounding unchanged regions more complex, thereby limiting the contextual discrimination capability of models for real change regions. Meanwhile, cropland change regions usually have large variations in scale, irregular shapes, and dispersed spatial distributions, which can easily lead to missed detections of small changes and incomplete extraction of local regions.

To address the above problems, a Global-Local Multiscale Feature Refinement Mamba Network (GMFRMNet) is proposed for cropland CD. GMFRMNet adopts a weight-shared Siamese Visual Mamba (VMamba) as the encoder. In addition, a Global-Local Context State-Space Block (GLCSB) and a Multiscale Channel-Spatial Refinement Block (MCSRB) are designed to enhance the contextual discrimination capability for change regions under complex conditions and the extraction capability for small and irregular change regions, respectively. Thus, missed detections and incomplete extraction of local regions are alleviated. The main contributions of this paper are summarized as follows.

(1) A Global-Local Multiscale Feature Refinement Mamba Network, namely GMFRMNet, is proposed for cropland CD. GMFRMNet adopts a weight-shared Siamese VMamba as the encoder and combines global-local context modeling with multiscale feature refinement to enhance the recognition and extraction of complex cropland change regions.

(2) A GLCSB is designed to address the insufficient interaction of change features and the limited contextual discrimination capability for change regions under complex conditions. Through a global relation modeling branch and a local context enhancement branch, GLCSB enhances the interaction of bi-temporal change features and improves the discrimination capability for true changed regions.

(3) An MCSRB is designed to address missed detections of small changes and incomplete extraction of local regions caused by large variations in scale, irregular shapes, and dispersed spatial distributions of cropland change regions. Through multiscale convolution branches, channel feature enhancement and spatial position refinement, MCSRB improves the extraction capability of the model for small and irregular cropland change regions.

The rest of this paper is organized as follows. Section 2 describes the structure of GMFRMNet in detail. Section 3 presents the experimental settings and analyzes the experimental results. Section 4 discusses the ablation experiments and model performance. Finally, Section 5 concludes this paper.

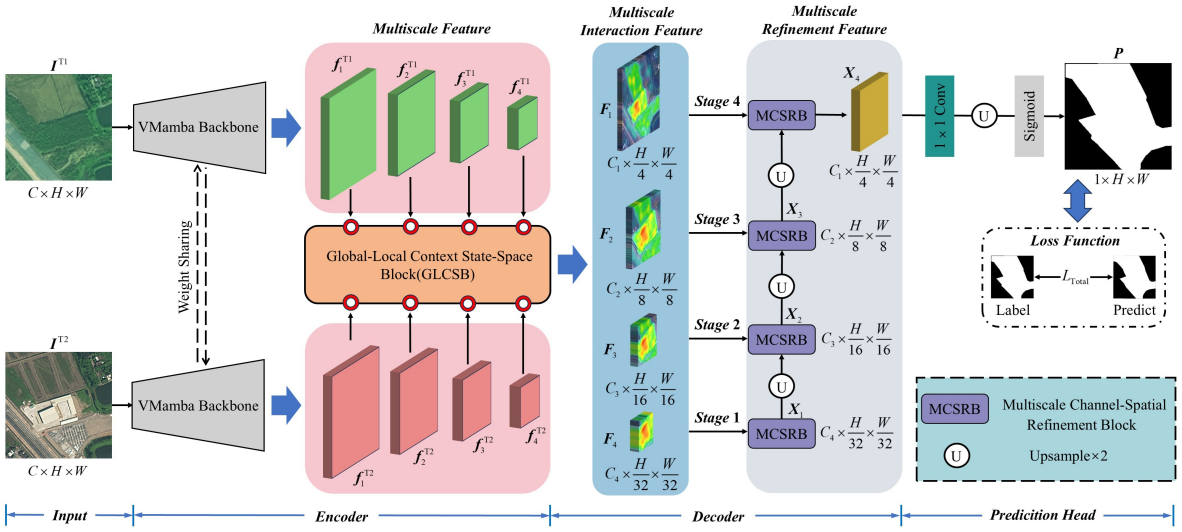


Figure 1: Overall architecture of GMFRMNet.

2. Methodology

2.1. Overall Architecture

The proposed GMFRMNet adopts an encoder-decoder architecture, as shown in Fig. 1. It mainly consists of a weight-shared Siamese VMamba encoder, GLCSB, MCSRB and a prediction head. The input of the network is a pair of bi-temporal images, denoted as $\boldsymbol{I}^{\mathrm{T}i} \in \mathbb{R}^{C\times H\times W}, i \in \{1,2\}$. For the image at the ith temporal phase, the weight-shared Siamese VMamba encoder outputs the feature $\boldsymbol{f}_n^{\mathrm{T}i}$ at the nth encoding layer, where $n \in \{1,2,3,4\}$. The bi-temporal multiscale features generated by the encoder are denoted as $\{\boldsymbol{f}_n^{\mathrm{T1}}, \boldsymbol{f}_n^{\mathrm{T2}}\}_{n=1}^{4}$, which provide multiscale feature representations from shallow to deep layers for the subsequent decoder.

In the encoding stage, GLCSB is introduced for context modeling and interaction of bi-temporal multiscale features. This module enhances the representation of local structural details while capturing long-range global dependencies. Thus, the problems of insufficient interaction between bi-temporal change features and limited contextual discrimination capability for change regions under complex temporal conditions are alleviated. Then, multiscale interaction features $\boldsymbol{F}_n$ are generated.

In the decoding stage, MCSRB is employed to progressively refine and fuse the interaction features from low-resolution deep layers to high-resolution shallow layers. As a result, multiscale refined features $\boldsymbol{X}_n$ are obtained. Finally, the change result map $\boldsymbol{P}$ is generated through the prediction head.

2.2. Weight-Shared Siamese VMamba Encoder

GMFRMNet employs a weight-shared Siamese VMamba encoder to extract multiscale features from bi-temporal images. For the input images $\boldsymbol{I}^{\mathrm{T}i}$, the two branches adopt the same network structure and share the same parameters for feature extraction, which can be expressed as,

$$\left\{\boldsymbol{f}_n^{\mathrm{T}i}\right\}_{n=1}^{4} = \mathrm{E}_\theta\left(\boldsymbol{I}^{\mathrm{T}i}\right) \tag{1}$$

where $\mathrm{E}_\theta(\cdot)$ denotes the VMamba encoder and θ represents the encoder parameters. Weight sharing enables the bi-temporal images to be represented in the same feature mapping space, avoiding feature shifts caused by parameter differences between the two branches.

As shown in Fig. 2(a), the VMamba encoder consists of four encoding layers, and the feature extraction unit in each layer is the visual state space block (VSS Block). First, the input image is transformed into initial features by the patch embedding layer. Then, semantic features at different levels are progressively extracted through four encoding layers and downsampling operations. The four features $\boldsymbol{f}_1^{\mathrm{T}i}, \boldsymbol{f}_2^{\mathrm{T}i}, \boldsymbol{f}_3^{\mathrm{T}i}$ and $\boldsymbol{f}_4^{\mathrm{T}i}$ output by the encoding layers correspond to $1/4$, $1/8$, $1/16$ and $1/32$ of the input image size, respectively.

As shown in Fig. 2(b), the VSS Block transforms the input features through layer normalization and linear projection. Depth-wise convolution is used to enhance local neighborhood information, and the 2D Selective Scan (2DSS) module is employed to perform sequential modeling on spatial features, so as to capture long-range spatial dependencies. Finally, the output features are obtained through linear projection and residual connection.

The 2DSS module is the core component of the VSS Block, and its sequential modeling process is shown in Fig. 2(c). Given the input feature $\boldsymbol{U}$ of the 2DSS module, $\boldsymbol{U}$ is unfolded into one-dimensional feature sequences $\boldsymbol{x}_d$ along four directions, where $d \in \{1,2,3,4\}$. This process is defined as,

$$\boldsymbol{x}_d = \mathrm{Scan}_d\left(\boldsymbol{U}\right) \tag{2}$$

where $\mathrm{Scan}_d(\cdot)$ denotes the scanning operation along the dth direction. The scanned feature $\boldsymbol{x}_d$ is then fed into the

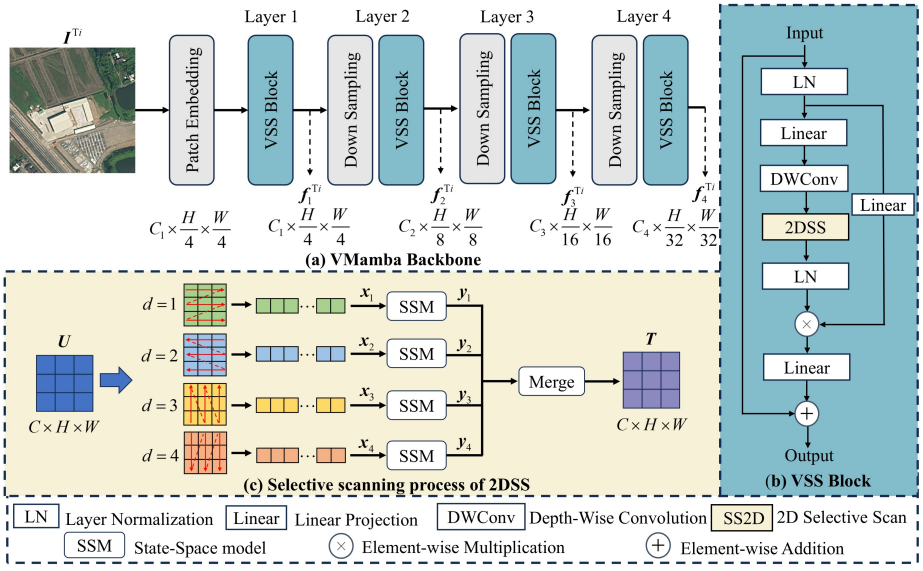


Figure 2: Architecture of the VMamba encoder. (a) VMamba backbone. (b) VSS Block. (c) Selective scanning process of 2DSS.

State-Space Model (SSM) for sequential context modeling to obtain the output sequence $\boldsymbol{y}_d$,

$$\boldsymbol{y}_d = \mathrm{SSM}_d\left(\boldsymbol{x}_d\right) \tag{3}$$

where $\mathrm{SSM}_d(\cdot)$ denotes the sequential modeling operation of the state space model in the dth direction.

The core of SSM is to transmit spatial context information across different positions in the sequence through recursive updates of hidden states. Let the input feature sequence, hidden state and output feature sequence at the tth position be $\boldsymbol{x}_t$, $\boldsymbol{h}_t$ and $\boldsymbol{y}_t$, respectively. For simplicity, the SSM process can be described in a general recurrent form as,

$$\begin{aligned} \boldsymbol{h}_t &= \boldsymbol{A}\boldsymbol{h}_{t-1} + \boldsymbol{B}\boldsymbol{x}_t \\ \boldsymbol{y}_t &= \boldsymbol{V}\boldsymbol{h}_t \end{aligned} \tag{4}$$

where $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{V}$ denote the state transition parameter, input mapping parameter and output mapping parameter, respectively. These parameters are adaptively learned during network training. Through the recursive update of the hidden state $\boldsymbol{h}_t$, SSM transmits long-range spatial information in long sequences, thereby capturing long-range spatial dependencies in feature maps.

Finally, the output sequences from the four directions are fused and reshaped into the two-dimensional feature $\boldsymbol{T}$,

$$\boldsymbol{T} = \mathrm{Merge}\left(\boldsymbol{y}_1, \boldsymbol{y}_2, \boldsymbol{y}_3, \boldsymbol{y}_4\right) \tag{5}$$

where $\mathrm{Merge}(\cdot)$ denotes sequence fusion and reconstruction. Through the above process, the weight-shared Siamese VMamba encoder extracts multiscale features from bi-temporal images in a unified feature space and enhances the ability to model long-range spatial relationships, providing rich spatial feature information for subsequent change feature interaction and feature refinement.

2.3. Global-Local Context State-Space Block (GLCSB)

In cropland CD, bi-temporal images not only contain semantic differences caused by land cover changes, but are also easily affected by phenological differences, illumination changes, shadows and differences in imaging conditions. These factors make the relationships between change regions and surrounding unchanged regions more complex. If only simple differencing or local feature fusion is used, it is difficult for the model to sufficiently model the global contextual relationships and local structural differences between bi-temporal images. This leads to insufficient interaction of bi-temporal change features and further limits the discrimination capability of the model for change regions. To address this problem, GLCSB is designed to improve the discrimination capability for true changes through global context modeling and local structure enhancement. The structure of GLCSB is shown in Fig. 3.

GLCSB jointly learns bi-temporal change information through a lightweight global relation modeling branch and an SSM local context enhancement branch. The global branch models the correlations among different spatial positions to enhance the global contextual discrimination capability for change regions. The local branch uses SSM and convolution operations to strengthen local texture and structural information, thereby improving the recognition capability of the model for cropland change regions under complex temporal conditions. Let the input bi-temporal features be $\boldsymbol{f}^{\mathrm{T}i}$. They are concatenated along the channel dimension to obtain the interaction feature $\boldsymbol{D}$. Different from feature differencing, this operation preserves the complete semantic information of the two temporal phases, enabling the network to adaptively learn change patterns according to contextual relationships. The features are then

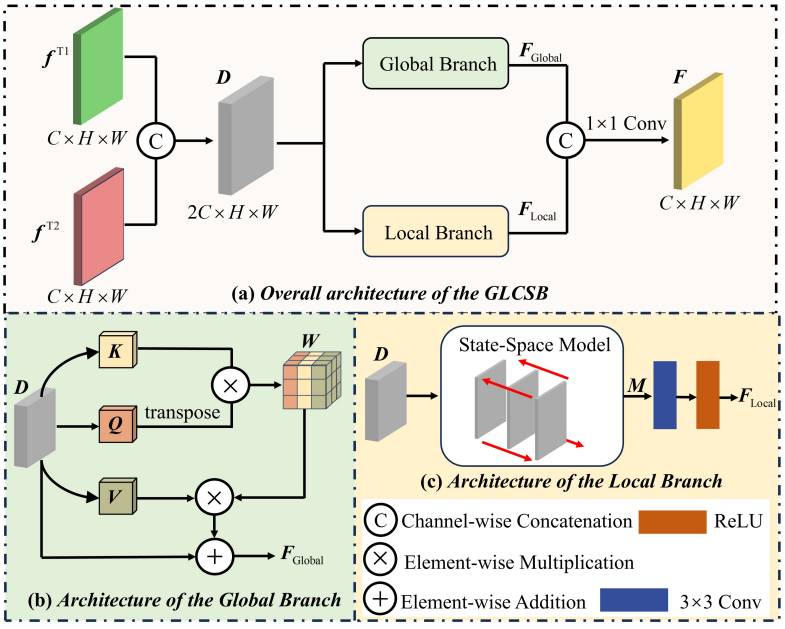


Figure 3: Architecture of GLCSB. (a) Overall architecture. (b) Global branch. (c) Local branch.

fed into the global branch and the local branch, as shown in Fig. 3(b) and Fig. 3(c), respectively.

The global branch is used to capture long-range spatial dependencies in the bi-temporal interaction features. To obtain query, key, and value representations for global relation modeling, the input feature $\boldsymbol{D}$ is mapped into three features, namely $\boldsymbol{Q}$, $\boldsymbol{K}$ and $\boldsymbol{V}$, through lightweight projections,

$$\begin{aligned} \boldsymbol{Q} &= \phi_{\mathrm{q}}(\boldsymbol{D}) \\ \boldsymbol{K} &= \phi_{\mathrm{k}}(\boldsymbol{D}) \\ \boldsymbol{V} &= \phi_{\mathrm{v}}(\boldsymbol{D}) \end{aligned} \tag{6}$$

where $\phi_{\mathrm{q}}(\cdot)$, $\phi_{\mathrm{k}}(\cdot)$ and $\phi_{\mathrm{v}}(\cdot)$ denote the feature mapping operations used to generate $\boldsymbol{Q}$, $\boldsymbol{K}$ and $\boldsymbol{V}$, respectively. The global weight $\boldsymbol{W}$ is obtained by calculating the correlation between $\boldsymbol{Q}$ and $\boldsymbol{K}$, which represents the global dependency relationships among different spatial positions. Then, $\boldsymbol{W}$ is multiplied by $\boldsymbol{V}$ to aggregate global contextual information, and a residual connection with $\boldsymbol{D}$ is adopted to obtain the global context interaction feature $\boldsymbol{F}_{\mathrm{Global}}$,

$$\begin{aligned} \boldsymbol{W} &= \mathrm{Softmax}\left(\boldsymbol{Q}\boldsymbol{K}^{\top}\right) \\ \boldsymbol{F}_{\mathrm{Global}} &= \boldsymbol{W} \cdot \boldsymbol{V} \oplus \boldsymbol{D} \end{aligned} \tag{7}$$

where $\mathrm{Softmax}(\cdot)$ denotes the Softmax function and $\oplus$ denotes element-wise addition. Through this process, long-range spatial contextual information is introduced while the original bi-temporal interaction features are retained, thereby enhancing the discrimination capability for true change regions.

The local branch is used to enhance local structural and texture information. The feature $\boldsymbol{D}$ is input into the SSM for context modeling to obtain the context-enhanced feature $\boldsymbol{M}$, followed by a 3×3 convolution and the ReLU activation function to further enhance local neighborhood information, producing the local context interaction feature $\boldsymbol{F}_{\mathrm{Local}}$,

$$\begin{aligned} \boldsymbol{M} &= \mathrm{SSM}(\boldsymbol{D}) \\ \boldsymbol{F}_{\mathrm{Local}} &= \mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}(\boldsymbol{M})\right) \end{aligned} \tag{8}$$

where $\mathrm{Conv}_{r\times r}(\cdot)$ denotes a convolution operation with a kernel size of $r\times r$ and $\mathrm{ReLU}(\cdot)$ denotes the ReLU activation function.

Finally, $\boldsymbol{F}_{\mathrm{Global}}$ and $\boldsymbol{F}_{\mathrm{Local}}$ are concatenated along the channel dimension, and a 1×1 convolution is used for feature fusion and channel compression to obtain the multiscale interaction feature $\boldsymbol{F}$,

$$\boldsymbol{F} = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(\boldsymbol{F}_{\mathrm{Global}}, \boldsymbol{F}_{\mathrm{Local}}\right)\right) \tag{9}$$

where $\mathrm{Concat}(\cdot)$ denotes feature concatenation. Through the above process, GLCSB integrates global contextual relationships and local structural information while preserving the complete semantic information of the bi-temporal features. Thus, change features are modeled from both global spatial relationships and local region structures, enhancing the contextual discrimination capability for cropland change regions under complex temporal conditions.

2.4. Multiscale Channel-Spatial Refinement Block (MCSRB)

In cropland CD, change regions usually show obvious differences in scale and spatial morphology. If only single-scale features are used for change region extraction, it is difficult to simultaneously consider global semantic information and local spatial information, which may lead to missed detections of small change regions and incomplete extraction of local change regions. Therefore, MCSRB

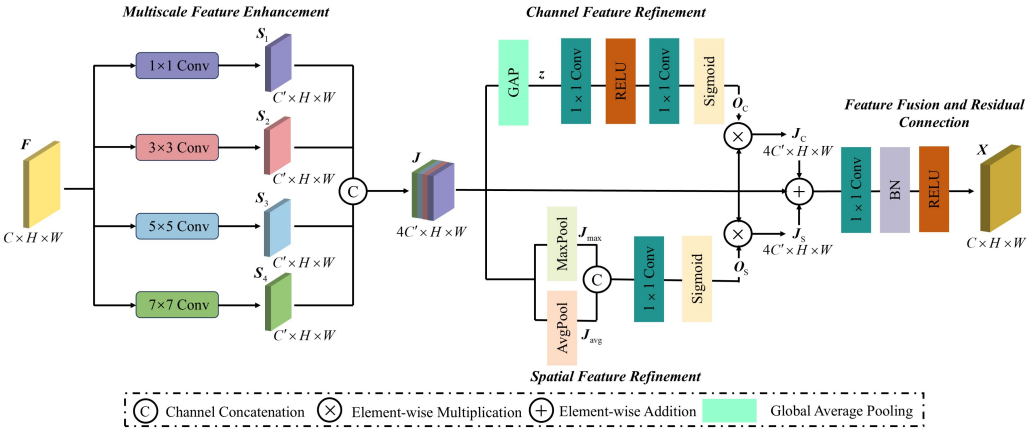


Figure 4: Architecture of MCSRB.

is proposed for multiscale feature enhancement and joint channel-spatial refinement, as shown in Fig. 4.

MCSRB extracts multiscale contextual features through four convolution branches with different kernel sizes to achieve multiscale feature enhancement. Specifically, the 1×1 convolution is used for channel information enhancement, the 3×3 convolution is used to extract local texture and spatial detail information, and the 5×5 and 7×7 convolutions are used to extract spatial information over larger ranges. The multiscale features $\boldsymbol{S}_n$ output by the four branches are concatenated along the channel dimension to obtain the multiscale fusion feature $\boldsymbol{J}$. This feature integrates spatial contextual information at different scales and can enhance the representation capability for both large change regions and small-scale change regions,

$$\begin{aligned} \boldsymbol{S}_1 &= \mathrm{Conv}_{1\times1}(\boldsymbol{F}) \\ \boldsymbol{S}_2 &= \mathrm{Conv}_{3\times3}(\boldsymbol{F}) \\ \boldsymbol{S}_3 &= \mathrm{Conv}_{5\times5}(\boldsymbol{F}) \\ \boldsymbol{S}_4 &= \mathrm{Conv}_{7\times7}(\boldsymbol{F}) \\ \boldsymbol{J} &= \mathrm{Concat}\left(\boldsymbol{S}_1, \boldsymbol{S}_2, \boldsymbol{S}_3, \boldsymbol{S}_4\right) \end{aligned} \tag{10}$$

To further highlight change information, MCSRB contains a channel refinement branch and a spatial feature refinement branch to enhance the fused feature $\boldsymbol{J}$ from the channel and spatial dimensions. In the channel refinement branch, global average pooling is first performed on $\boldsymbol{J}$ to obtain the channel descriptor $\boldsymbol{z}$. Then, a 1×1 convolution, the ReLU activation function, another 1×1 convolution and the Sigmoid function are sequentially used to generate the channel attention weight $\boldsymbol{O}_\mathrm{C}$. The channel attention weight is multiplied element-wise with the original feature $\boldsymbol{J}$ to obtain the channel-enhanced feature $\boldsymbol{J}_\mathrm{C}$,

$$\begin{aligned} \boldsymbol{z} &= \mathrm{GAP}(\boldsymbol{J}) \\ \boldsymbol{O}_\mathrm{C} &= \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{1\times1}(\boldsymbol{z})\right)\right)\right) \\ \boldsymbol{J}_\mathrm{C} &= \boldsymbol{O}_\mathrm{C} \otimes \boldsymbol{J} \end{aligned} \tag{11}$$

where $\mathrm{GAP}(\cdot)$ denotes global average pooling, $\sigma(\cdot)$ denotes the Sigmoid function and $\otimes$ denotes element-wise multiplication. This process adaptively enhances the channel responses that are more sensitive to cropland change regions while suppressing redundant background information.

In the spatial feature refinement branch, MCSRB further enhances the location representation and spatial response capability of change regions. Channel-wise max pooling and channel-wise average pooling are performed on feature $\boldsymbol{J}$, respectively. The output results are concatenated along the channel dimension and then passed through a 1×1 convolution and the Sigmoid function to generate the spatial attention weight $\boldsymbol{O}_\mathrm{S}$. Spatial weighted enhancement is then performed on feature $\boldsymbol{J}$ to obtain the spatial-enhanced feature $\boldsymbol{J}_\mathrm{S}$,

$$\begin{aligned} \boldsymbol{J}_{\max} &= \mathrm{MaxPool}(\boldsymbol{J}) \\ \boldsymbol{J}_{\mathrm{avg}} &= \mathrm{AvgPool}(\boldsymbol{J}) \\ \boldsymbol{O}_\mathrm{S} &= \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(\boldsymbol{J}_{\max}, \boldsymbol{J}_{\mathrm{avg}}\right)\right)\right) \\ \boldsymbol{J}_\mathrm{S} &= \boldsymbol{O}_\mathrm{S} \otimes \boldsymbol{J} \end{aligned} \tag{12}$$

The spatial refinement branch guides the network to focus more on the locations and spatial distributions of change regions, thereby reducing false alarms in complex background regions and improving the recognition performance for small and irregular change regions. Finally, MCSRB fuses the original multiscale fusion feature, the channel-enhanced feature and the spatial-enhanced feature to obtain the refined feature $\boldsymbol{X}$,

$$\boldsymbol{X} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\boldsymbol{J} + \boldsymbol{J}_\mathrm{C} + \boldsymbol{J}_\mathrm{S}\right)\right)\right) \tag{13}$$

where $\mathrm{BN}(\cdot)$ denotes batch normalization. Through the above feature fusion strategy, MCSRB further integrates channel discrimination information and spatial location information while preserving multiscale contextual information, thereby obtaining a more refined representation of change features.

2.5. Adaptive Weighted Loss Function

In CD tasks, severe class imbalance often exists because the number of unchanged pixels in the training data is much larger than that of changed pixels. This class imbalance may cause the network to focus more on unchanged samples during training while ignoring changed samples. To address this problem, this paper combines binary cross-entropy (BCE) loss and Dice loss for objective optimization. BCE loss focuses on pixel-level classification accuracy, while Dice loss improves the regional overlap between predicted and ground-truth change regions. The BCE and Dice loss functions are formulated as,

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[g_i \ln\left(p_i\right) + \left(1-g_i\right)\ln\left(1-p_i\right)\right] \quad (14)$$

$$L_{\mathrm{Dice}} = 1 - \frac{2\sum_{i=1}^{N} g_i p_i + \varepsilon}{\sum_{i=1}^{N} g_i + \sum_{i=1}^{N} p_i + \varepsilon} \quad (15)$$

where N denotes the total number of pixels, g_i denotes the ground truth change label of the ith pixel, $g_i \in \{0, 1\}$ and p_i denotes the predicted probability that the ith pixel belongs to the changed class, $p_i \in (0, 1)$. In this paper, ε is set to 1×10^{-5} to avoid division by zero. The total loss function is expressed as,

$$L_{\mathrm{Total}} = \alpha L_{\mathrm{BCE}} + \beta L_{\mathrm{Dice}} \quad (16)$$

where α and β denote the weights of the BCE loss and Dice loss, respectively. Since fixed weights are difficult to adapt to the different contributions of the two losses at different training stages and require complex manual tuning, an adaptive weight adjustment strategy is adopted for α and β. The weights are adaptively adjusted according to the relative magnitudes of the two losses during training, enabling the model to pay more attention to the loss term with higher current optimization difficulty. The weights are defined as follows,

$$\alpha = \frac{L_{\mathrm{BCE}}}{L_{\mathrm{BCE}} + L_{\mathrm{Dice}}}, \quad \beta = \frac{L_{\mathrm{Dice}}}{L_{\mathrm{BCE}} + L_{\mathrm{Dice}}} \quad (17)$$

3. Experiments and Results

3.1. Experimental Datasets

(1) CropLand Change Detection (CLCD) dataset (Liu et al., 2022) contains 600 pairs of bi-temporal cropland change images with a size of 512 × 512, including 360 pairs for training, 120 pairs for validation and 120 pairs for testing. The bi-temporal images are derived from Gaofen-2 remote sensing images of Guangdong Province acquired from 2017 to 2019, with a spatial resolution of 0.5-2 m. To adapt to the input size of the proposed model, the original dataset is cropped into images with a size of 256 × 256.

(2) Jilin-1 (JL-1) dataset (Wu et al., 2025) is derived from Jilin-1 satellite remote sensing images, with a spatial resolution better than 0.75 m. It contains three bands, namely R, G and B. The dataset consists of 10,000 pairs of bi-temporal images with a size of 256 × 256, including 6,000 pairs for training, 2,000 pairs for validation and 2,000 pairs for testing. This dataset contains various types of cropland changes, including cropland converted to roads, buildings, forest and grassland and other land types, as well as roads, buildings, forest and grassland and other land types converted to cropland. It also includes non-cropland change regions. Since this paper focuses on the extraction of cropland change regions, the original dataset is reannotated. Non-cropland change regions are labeled as 0, while cropland change regions are labeled as 1.

(3) Peixian Cultivated Land Change Detection (PX-CLCD) dataset (Miao et al., 2024) consists of 5,170 pairs of bi-temporal images with a spatial resolution of 1 m and an image size of 256 × 256 pixels. The dataset is divided into training, validation and testing sets at a ratio of 6:2:2. PX-CLCD contains cropland change images in Peixian County from 2018 to 2021. The main change types include cropland converted to greenhouses, roads, artificial lakes and buildings.

3.2. Experimental Setup and Evaluation Metrics

All experiments in this study are implemented based on the PyTorch framework and trained on an NVIDIA RTX 3090 GPU with 24 GB of memory. All models are trained using the Adam optimizer. The initial learning rate is set to 1×10^{-4}, the weight decay coefficient is set to 5×10^{-3}, the batch size is set to 16 and the number of training epochs is set to 100.

To verify the effectiveness and superiority of the proposed method, eight representative CD methods are selected for comparative experiments, including FC-EF (Daudt et al., 2018), SNUNet (Fang et al., 2022), BIT (Chen et al., 2022), ChangeFormer (Bandara and Patel, 2022), MSCANet (Liu et al., 2022), FTA-Net (Zhu et al., 2025), ChangeMamba (Chen et al., 2024a) and CDMamba (Zhang et al., 2025). To ensure fair comparison, the network architectures and main modules of the comparison methods follow their original papers or open-source implementations, while all methods are trained and tested under the same data splits, input size, optimizer settings, training epochs and experimental environment.

To comprehensively evaluate model performance, Overall Accuracy (OA), Intersection over Union (IoU), Precision (Pre), Recall (Rec) and F1-Score (F1) are selected as evaluation metrics (Liu et al., 2022; Wang et al., 2026; Zhu et al., 2025). Among them, F1 and IoU comprehensively reflect the detection accuracy and spatial overlap degree of change regions; therefore, they are used as the core metrics for evaluating CD performance.

Table 1
Quantitative results on the CLCD, JL-1 and PX-CLCD datasets. **Bold** and underline indicate the best and second-best results, respectively.

Dataset	Metric	FC-EF	SNUNet	BIT	ChangeFormer	MSCANet	FTA-Net	ChangeMamba	CDMamba	GMFRMNet
	FLOPs(G)	3.58	54.83	15.75	21.18	59.08	6.64	25.75	59.83	30.46
	Params(M)	1.35	50.44	17.99	29.75	16.42	28.90	42.93	21.87	26.15
CLCD	OA(%)	94.57	93.88	95.23	94.86	95.29	95.84	94.90	94.51	**97.43**
	IoU(%)	46.74	39.95	51.44	50.25	50.53	53.99	48.31	45.97	**68.99**
	F1(%)	63.71	57.09	67.93	66.89	67.14	70.12	65.15	62.98	**81.65**
	Pre(%)	63.38	59.67	67.92	64.22	69.79	75.23	66.24	63.22	**87.17**
	Rec(%)	64.03	54.72	67.95	69.79	64.68	65.66	64.10	62.75	**76.79**
JL-1	OA(%)	94.84	94.56	95.65	95.22	95.56	95.88	95.55	95.11	**96.81**
	IoU(%)	53.94	45.37	60.10	57.77	58.55	62.44	60.89	56.18	**69.82**
	F1(%)	70.08	62.42	75.08	73.24	73.86	76.87	75.69	71.95	**82.23**
	Pre(%)	76.86	**88.83**	79.41	77.18	82.48	81.29	77.66	78.03	86.15
	Rec(%)	64.39	48.12	71.19	69.68	66.87	72.92	73.82	66.74	**78.65**
PX-CLCD	OA(%)	95.96	92.96	93.85	98.65	98.30	97.54	98.77	98.45	**99.12**
	IoU(%)	72.52	75.81	80.12	90.07	87.68	82.08	90.88	88.60	**93.42**
	F1(%)	84.07	86.24	88.96	94.78	93.43	90.16	95.22	93.96	**96.60**
	Pre(%)	85.66	88.00	91.22	94.40	92.87	93.07	95.64	94.86	**96.57**
	Rec(%)	82.54	84.55	86.83	95.07	94.01	87.42	94.81	93.07	**96.63**

3.3. Experimental Results and Analysis

3.3.1. Quantitative Evaluation

The quantitative evaluation results of different methods on the CLCD, JL-1 and PX-CLCD datasets are shown in Table 1. Overall, GMFRMNet outperforms the comparison methods in the main metrics on all three datasets, especially achieving clear improvements in the two core metrics, F1 and IoU. This indicates that GMFRMNet can accurately extract cropland change regions and achieve better spatial overlap.

On the CLCD dataset, all evaluation metrics of GMFRMNet are superior to those of the other comparison methods. Compared with the second-best method, GMFRMNet improves OA, IoU, F1, Pre, and Rec by 1.59, 15.00, 11.53, 11.94, and 7.00 percentage points, respectively. In particular, the improvements in F1 and IoU are significant, indicating that GMFRMNet has obvious advantages in detection accuracy and regional overlap.

On the JL-1 dataset, GMFRMNet achieves the best results in OA, IoU, F1 and Rec, reaching 96.81%, 69.82%, 82.23% and 78.65%, respectively. Compared with the second-best method, GMFRMNet improves OA, IoU, F1 and Rec by 0.93, 7.38, 5.36 and 4.83 percentage points, respectively. Although SNUNet obtains the highest Pre of 88.83%, its Rec is only 48.12%, indicating that SNUNet produces fewer false alarms but suffers from more missed detections. In contrast, GMFRMNet achieves Pre and Rec values of 86.15% and 78.65%, respectively, showing a better balance between the two metrics. This demonstrates that GMFRMNet effectively reduces missed detections while maintaining high detection accuracy.

On the PX-CLCD dataset, GMFRMNet outperforms the comparison methods in all five metrics. The OA, IoU, F1, Pre and Rec reach 99.12%, 93.42%, 96.60%, 96.57% and 96.63%, respectively. Compared with the second-best method, GMFRMNet improves these metrics by 0.35, 2.54, 1.38, 0.93 and 1.56 percentage points, respectively. Although the comparison methods generally achieve high accuracy on the PX-CLCD dataset, GMFRMNet still maintains leading performance across all metrics.

3.3.2. Qualitative Evaluation

The visualization results of different methods on the CLCD, JL-1 and PX-CLCD datasets are shown in Figs. 5–7. In these figures, white represents correctly detected change regions, black represents unchanged regions, red represents false alarm regions and green represents missed detection regions.

On the CLCD dataset, as shown in Fig. 5, FC-EF, SNUNet, BIT and ChangeFormer show obvious missed detections in large change regions, such as S1, S2 and S5. This indicates that these methods have insufficient overall representation capability for complex cropland changes. For the small change regions in S3 and the scattered changes in S4, the comparison methods suffer from local missed detections and incomplete region extraction. In contrast, GMFRMNet can extract change regions more completely. Especially for the long and narrow road change in S6, the proposed method can preserve the continuous shape of the change region more effectively, indicating that GMFRMNet has a strong representation capability for multiscale change features.

On the JL-1 dataset, as shown in Fig. 6, the comparison methods suffer from different degrees of false alarms and missed detections in S1, S2 and S3. This problem is particularly evident for SNUNet. Although SNUNet achieves

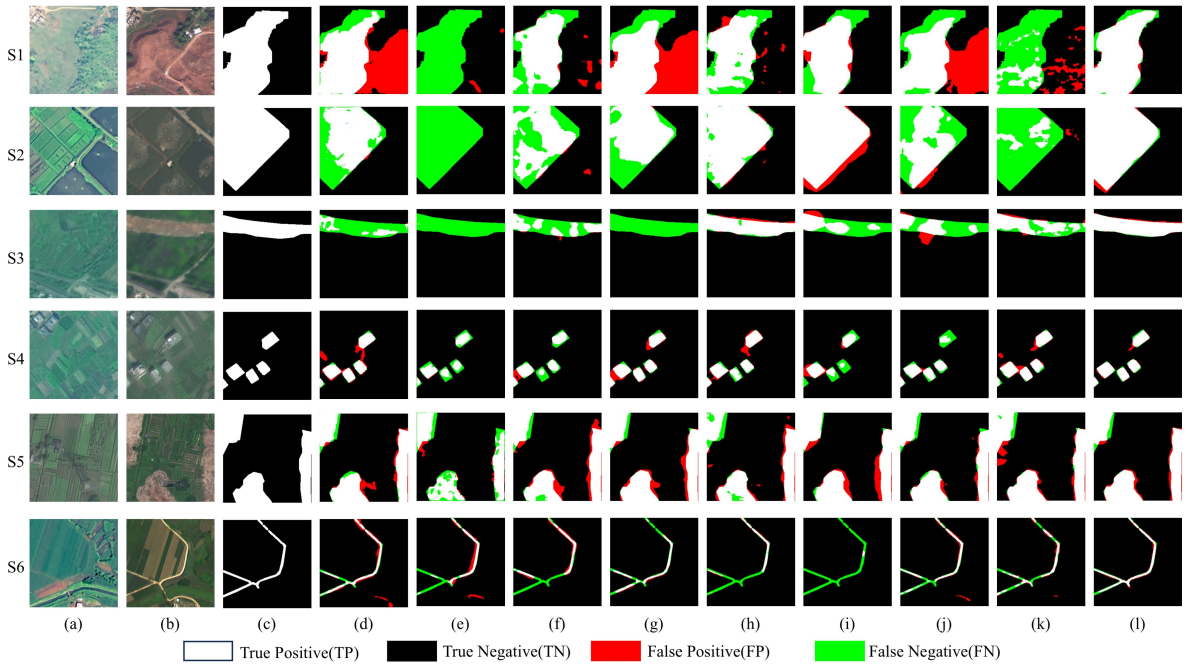


Figure 5: Visualization results on the CLCD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) FC-EF. (e) SNUNet. (f) BIT. (g) ChangeFormer. (h) MSCANet. (i) FTA-Net. (j) ChangeMamba. (k) CDMamba. (l) GMFRMNet. S1–S6 represent six representative examples.

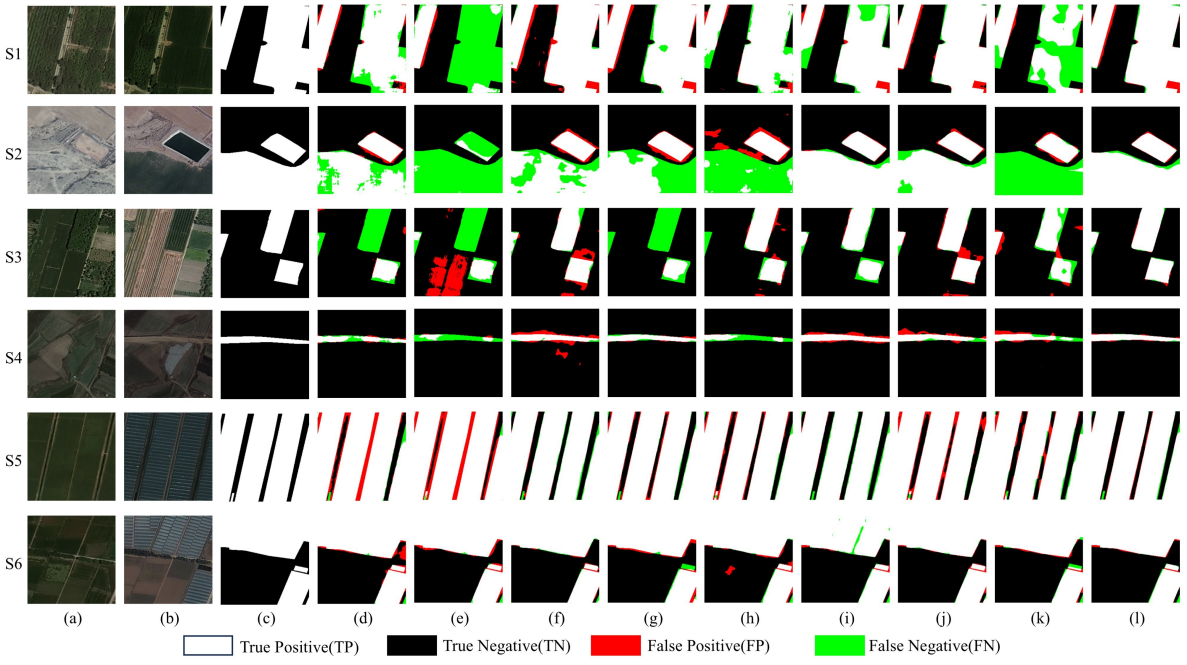


Figure 6: Visualization results on the JL-1 dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) FC-EF. (e) SNUNet. (f) BIT. (g) ChangeFormer. (h) MSCANet. (i) FTA-Net. (j) ChangeMamba. (k) CDMamba. (l) GMFRMNet. S1–S6 represent six representative examples.

a high Pre value, a large number of green missed detection regions appear in the visualization results, which is consistent with its low Rec value. In the long and narrow or morphologically complex change regions in S4, S5 and S6, methods such as FC-EF, SNUNet and MSCANet show obvious missed detections, false alarms and incomplete local regions. In contrast, the detection results of GMFRMNet are closer to the ground truth. In particular, GMFRMNet

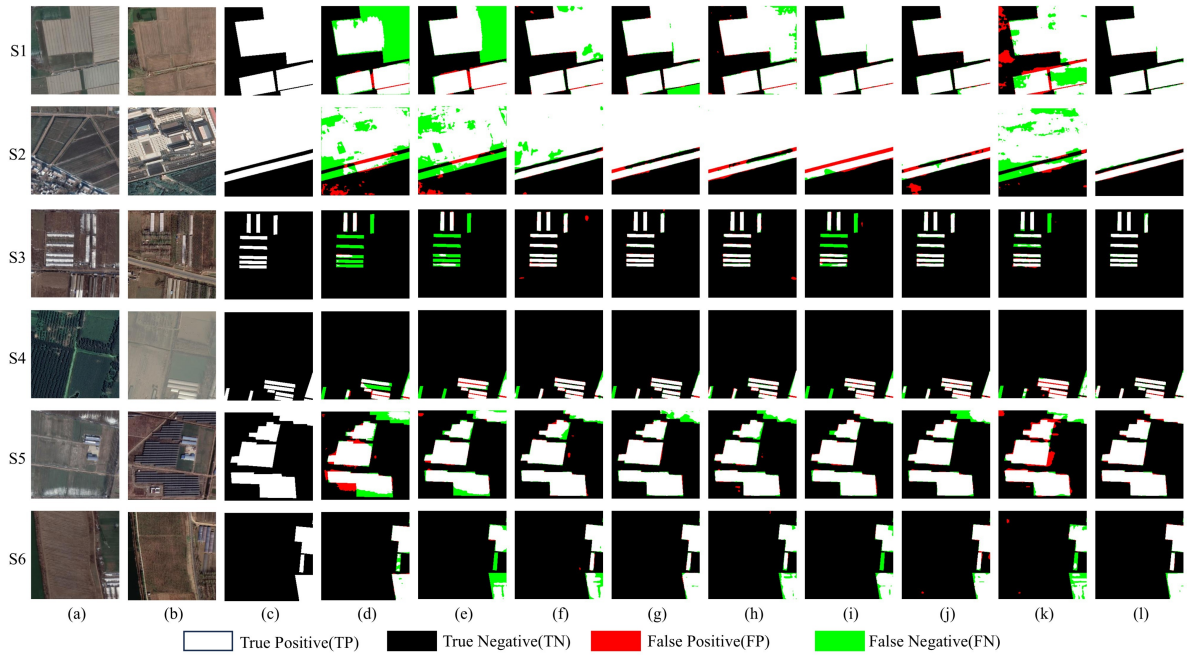


Figure 7: Visualization results on the PX-CLCD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) FC-EF. (e) SNUNet. (f) BIT. (g) ChangeFormer. (h) MSCANet. (i) FTA-Net. (j) ChangeMamba. (k) CDMamba. (l) GMFRMNet. S1–S6 represent six representative examples.

performs more stably in slender and complex change regions, such as roads and greenhouse facilities in S4 and S6, indicating that it can better maintain the integrity and continuity of change regions.

On the PX-CLCD dataset, as shown in Fig. 7, in large change regions, such as S1, S2 and S5, GMFRMNet produces detection results that are highly consistent with the ground truth and preserves the changed regions more completely. In small greenhouse change scenes, such as S3, S4 and S6, GMFRMNet can also extract changed regions more accurately. In comparison, FC-EF, SNUNet, BIT and ChangeFormer still show certain false alarms and local missed detections. Although ChangeMamba and CDMamba achieve relatively good overall performance, they still have a small number of local missing regions and detail omissions. Overall, the results of GMFRMNet are more accurate, indicating that it has a stronger capability to extract fine-scale changes in complex cropland change scenes.

The visualization results on the three datasets show that GMFRMNet can effectively reduce false alarms caused by complex background interference and missed detections of small change regions. It also maintains good detection performance for cropland change regions with different scales, shapes and spatial distributions. These results are consistent with the quantitative evaluation results, further verifying the effectiveness of the proposed GMFRMNet for cropland CD.

4. Discussion

4.1. Ablation Study

To verify the effectiveness of GLCSB and MCSRB in GMFRMNet, ablation experiments are conducted on the CLCD, JL-1 and PX-CLCD datasets. In these experiments, Base denotes the VMamba baseline network without GLCSB and MCSRB. Base+GLCSB denotes the baseline network with GLCSB. Base+MCSRB denotes the baseline network with MCSRB. GMFRMNet denotes the complete model with both modules. The quantitative results of the ablation experiments are shown in Table 2. Compared with Base, the separate introduction of GLCSB and MCSRB improves the model performance. Compared with the Base model, the complete GMFRMNet improves the F1 score by 1.21, 5.63, and 1.17 percentage points and the IoU by 1.70, 7.75, and 2.16 percentage points on the CLCD, JL-1, and PX-CLCD datasets, respectively. These results indicate that the best model performance is achieved when GLCSB and MCSRB are used together.

As shown in the visualization ablation results in Fig. 8, the Base model suffers from local false alarms, incomplete region extraction and missed detections of small changes on different datasets. S6 is a scene in PX-CLCD where cropland is converted into greenhouses. Due to phenological differences between seasons and the similar features of cropland and greenhouses, the Base model has insufficient contextual discrimination capability for bi-temporal change features under complex temporal conditions, causing some unchanged regions to be incorrectly detected as change

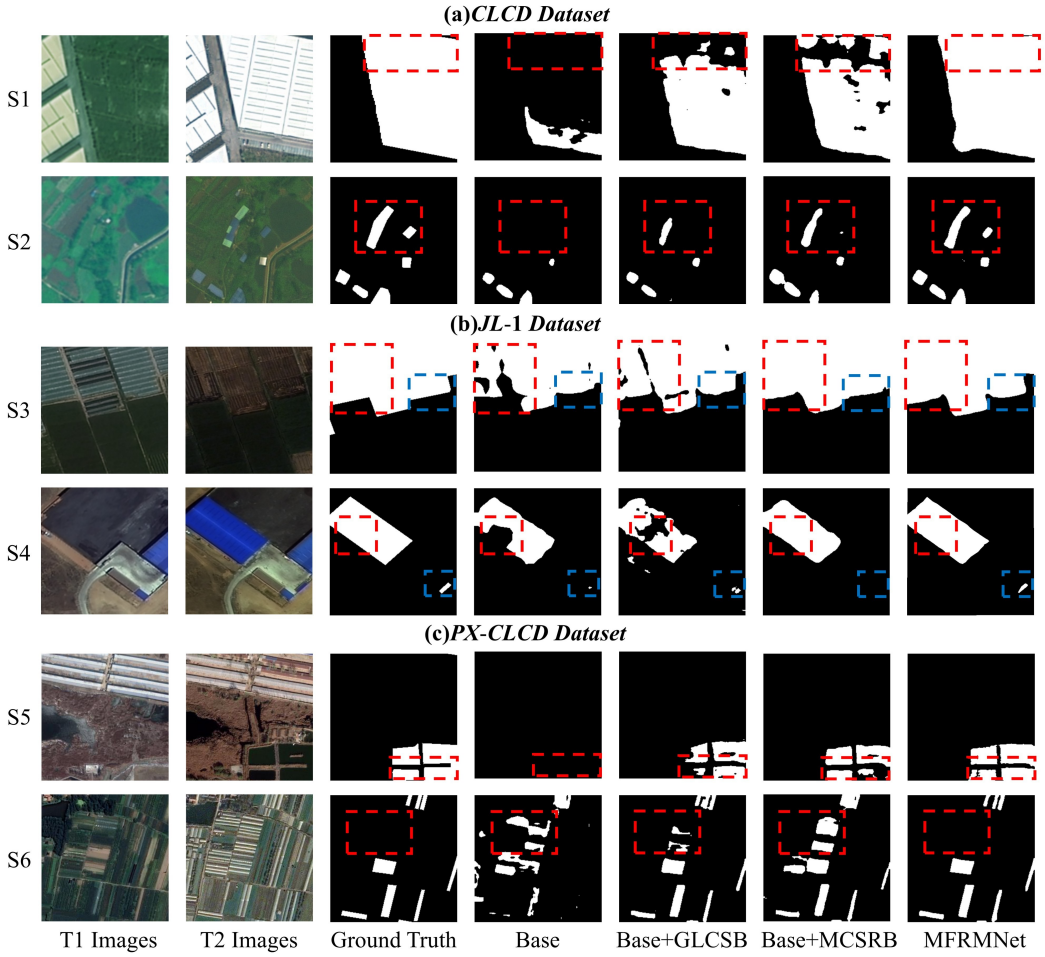


Figure 8: Visualization results of the ablation study. S1 and S2 are examples from CLCD dataset. S3 and S4 are examples from JL-1 dataset. S5 and S6 are from PX-CLCD dataset. The red and blue boxes highlight the regions of interest.

Table 2
Ablation results of GMFRMNet. **Bold** indicates the best result.

Method	CLCD		JL-1		PX-CLCD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
Base	80.44	67.29	76.60	62.07	95.43	91.26
Base+GLCSB	80.70	67.66	79.93	66.57	96.12	92.53
Base+MCSRB	81.52	68.81	80.74	67.70	96.21	92.70
GMFRMNet	**81.65**	**68.99**	**82.23**	**69.82**	**96.60**	**93.42**

regions. After GLCSB is introduced, the false alarm regions are reduced. This indicates that GLCSB improves the discrimination capability of the model for change regions by enhancing the interaction between global contextual relationships and local structural information. As shown in S2 and S3, after MCSRB is introduced, the model extracts small and irregular change regions more completely. This indicates that MCSRB can alleviate missed detections of small changes and incomplete extraction of irregular change regions. The prediction results of the complete GMFRMNet are the closest to the ground truth, with more complete change regions and fewer local missing regions. Overall, both the quantitative metrics and visualization results demonstrate that GLCSB and MCSRB can effectively improve model performance and their joint use further enhances the ability of the model to recognize cropland change regions.

4.2. Loss Weight Experiment

To verify the effect of the adaptive weighted loss function designed in this study, six experiments are conducted on the three datasets under the same experimental conditions while keeping the GMFRMNet network structure unchanged. In these experiments, a single BCE loss, a single Dice loss and combined loss functions with different weights are used for comparison. As shown in Table 3, when only a single loss function is used, the overall accuracy metrics are relatively low. In contrast, the adaptive weighted loss can dynamically coordinate the contributions of the two losses during training and achieves slightly better and more stable performance across the three datasets. Therefore, the adaptive weighted loss function is adopted to obtain relatively balanced performance across different datasets.

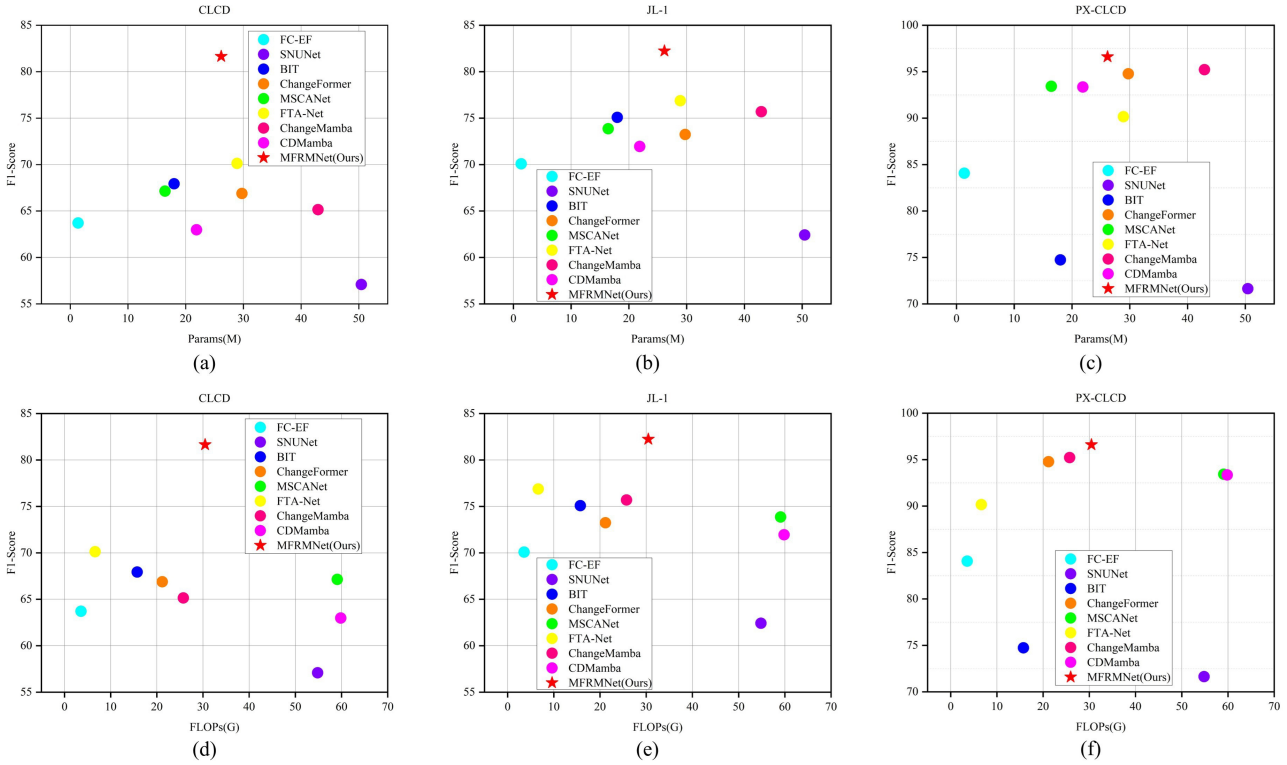


Figure 9: Comparison of model complexity and accuracy among different methods. (a) and (d) CLCD dataset. (b) and (e) JL-1 dataset. (c) and (f) PX-CLCD dataset.

Table 3
Comparison results of different loss weights. **Bold** and underline indicate the best and second-best results, respectively.

Weight		CLCD		JL-1		PX-CLCD	
α	β	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
1	0	80.74	67.70	81.49	68.76	95.75	91.85
0.3	0.7	81.13	68.25	81.75	69.13	96.56	93.35
0.5	0.5	81.58	68.89	82.10	69.74	96.13	92.55
0.7	0.3	80.79	67.77	82.03	69.38	95.79	91.92
0	1	80.63	67.56	81.92	69.43	95.72	91.80
Adaptive weights		**81.65**	**68.99**	**82.23**	**69.82**	**96.60**	**93.42**

4.3. Model Efficiency

To further evaluate the relationship between detection accuracy and model complexity, the number of parameters (Params) and floating-point operations (FLOPs) of each method are counted under the input of a pair of 256×256 remote sensing images. The specific results are shown in Table 1. Based on these results, scatter plots of model complexity and accuracy are constructed by combining the F1 scores on the three datasets, as shown in Fig. 9. In Fig. 9(a)-(c), the horizontal axis denotes Params, while in Fig. 9(d)-(f), the horizontal axis denotes FLOPs. In all subfigures, the vertical axis represents the F1 score. A smaller value on the horizontal axis indicates lower model complexity, while a higher value on the vertical axis indicates higher detection accuracy. Therefore, methods closer to the upper-left corner generally achieve better accuracy-efficiency trade-offs.

FC-EF has relatively low complexity, but its detection accuracy is insufficient. SNUNet and MSCANet have relatively high model complexity, but their performance gains are limited. Methods such as BIT, ChangeFormer and FTA-Net achieve a certain balance between accuracy and efficiency, but their detection accuracy is still lower than that of GMFRMNet. In contrast, GMFRMNet contains 26.15 M parameters and requires 30.46 G FLOPs, indicating moderate model complexity. Meanwhile, it achieves the highest F1 scores on all three datasets. These results indicate that GMFRMNet achieves high detection accuracy with moderate model complexity, demonstrating a favorable balance between accuracy and efficiency.

5. Conclusion

To address complex temporal differences, insufficient interaction of bi-temporal change features and the difficulty of extracting irregular change regions with large variations in scale in cropland CD, a global-local multiscale feature refinement Mamba network, named GMFRMNet, is proposed in this paper. GMFRMNet adopts a weight-shared Siamese VMamba as the encoder. GLCSB, MCSRB and an adaptive weighted loss function are designed to

enhance the discrimination and extraction capabilities for complex cropland change regions. The main conclusions are as follows.

(1) Experimental results on the CLCD, JL-1 and PX-CLCD datasets show that the proposed GMFRMNet achieves the best performance, with F1 scores of 81.65%, 82.23% and 96.60%, respectively. The visualization results further show that GMFRMNet can extract change regions more completely and maintain better spatial continuity in small-scale, elongated and complex cropland change scenes.

(2) The ablation experiments confirm the effectiveness of GLCSB and MCSRB. Compared with the Base model, GMFRMNet improves F1 by 1.21, 5.63 and 1.17 percentage points and IoU by 1.70, 7.75 and 2.16 percentage points on the CLCD, JL-1 and PX-CLCD datasets, respectively. The visualization results further show that GLCSB reduces false alarms, while MCSRB improves the extraction completeness of small, scattered and irregular cropland change regions.

(3) The loss weight experiment shows that the adaptive weighting strategy achieves slightly better and more stable performance across the three datasets. The model efficiency analysis indicates that GMFRMNet achieves high detection accuracy with moderate model complexity, demonstrating a favorable balance between accuracy and efficiency.

Although GMFRMNet achieves promising performance, it still has some limitations. The model retains moderate computational complexity, and its generalization ability under cross-region, cross-sensor, and different seasonal conditions requires further verification. In future work, we will optimize the network structure through lightweight feature extraction and efficient feature fusion, reduce redundant parameters, and incorporate multi-source remote sensing data to improve the robustness and adaptability of GMFRMNet in complex cropland change scenarios.

Declarations of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The CLCD, JL-1, and PX-CLCD datasets used in this study are publicly available at https://github.com/liumency/CropLand-CD, https://www.jl1mall.com, and https://github.com/niuzhan/Peixian-Cultivated-land-Change-detection-dataset, respectively.

Additional data generated during this study are available from the corresponding author upon reasonable request.

Author contribution

Shuhan Jia: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. Qi Zhao: Writing – original draft, Visualization, Validation, Conceptualization. Quanhua Zhao: Writing – review & editing, Methodology. Yu Li: Writing – review & editing, Methodology.

References

Bandara, W.G.C., Patel, V.M., 2022. A transformer-based siamese network for change detection, in: IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium, pp. 207–210. doi:10.1109/IGARSS46834.2022.9883686.

Celik, T., Ma, K.K., 2010. Unsupervised change detection for satellite images using dual-tree complex wavelet transform. IEEE Transactions on Geoscience and Remote Sensing 48, 1199–1210. doi:10.1109/TGRS.2009.2029095.

Chen, H., Qi, Z., Shi, Z., 2022. Remote sensing image change detection with transformers. IEEE Transactions on Geoscience and Remote Sensing 60, 1–14. doi:10.1109/TGRS.2021.3095166.

Chen, H., Song, J., Han, C., Xia, J., Yokoya, N., 2024a. Changemamba: Remote sensing change detection with spatiotemporal state space model. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 62. doi:10.1109/TGRS.2024.3417253.

Chen, H., Wu, C., Du, B., Zhang, L., 2019. Deep siamese multi-scale convolutional network for change detection in multi-temporal vhr images, in: 2019 10th International Workshop on the Analysis of Multitemporal Remote Sensing Images (MultiTemp), pp. 1–4. doi:10.1109/Multi-Temp.2019.8866947.

Chen, K., Chen, B., Liu, C., Li, W., Zou, Z., Shi, Z., 2024b. Rsmamba: Remote sensing image classification with state space model. IEEE GEOSCIENCE AND REMOTE SENSING LETTERS 21. doi:10.1109/LGRS.2024.3407111.

Daudt, R.C., Le Saux, B., Boulch, A., 2018. Fully convolutional siamese networks for change detection, in: 2018 25th IEEE International Conference on Image Processing (ICIP), pp. 4063–4067. doi:10.1109/ICIP.2018.8451652.

Ding, L., Hong, D., Zhao, M., Chen, H., Li, C., Deng, J., Yokoya, N., Bruzzone, L., Chanussot, J., 2025. A survey of sample-efficient deep learning for change detection in remote sensing: Tasks, strategies, and challenges. IEEE Geoscience and Remote Sensing Magazine 13, 164–189. doi:10.1109/MGRS.2025.3533605.

Fang, S., Li, K., Shao, J., Li, Z., 2022. Snunet-cd: A densely connected siamese network for change detection of vhr images. IEEE GEOSCIENCE AND REMOTE SENSING LETTERS 19. doi:10.1109/LGRS.2021.3056416.

Feng, W., Sui, H., Tu, J., Huang, W., Sun, K., 2018. A novel change detection approach based on visual saliency and random forest from multi-temporal high-resolution remote-sensing images. International Journal of Remote Sensing 39, 7998–8021. doi:10.1080/01431161.2018.1479794.

Gong, M., Cao, Y., Wu, Q., 2012. A neighborhood-based ratio approach for change detection in sar images. IEEE Geoscience and Remote Sensing Letters 9, 307–311. doi:10.1109/LGRS.2011.2167211.

Gong, M., Yang, H., Zhang, P., 2017. Feature learning and change feature classification based on deep learning for ternary change detection in sar images. ISPRS JOURNAL OF PHOTOGRAMMETRY AND

REMOTE SENSING 129, 212–225. doi:10.1016/j.isprsjprs.2017.05.001.

Guo, Q., Zhang, J., Zhu, S., Zhong, C., Zhang, Y., 2022. Deep multi-scale siamese network with parallel convolutional structure and self-attention for change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 60. doi:10.1109/TGRS.2021.3131993.

Hou, B., Wang, Y., Liu, Q., 2017. Change detection based on deep features and low rank. IEEE GEOSCIENCE AND REMOTE SENSING LETTERS 14, 2418–2422. doi:10.1109/LGRS.2017.2766840.

Kuncheva, L.I., Faithfull, W.J., 2014. Pca feature extraction for change detection in multidimensional unlabeled data. IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS 25, 69–80. doi:10.1109/TNNLS.2013.2248094.

Li, Q., Zhong, R., Du, X., Du, Y., 2022. Transunetcd: A hybrid transformer network for change detection in optical remote-sensing images. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 60. doi:10.1109/TGRS.2022.3169479.

Li, Z., Ouyang, B., Qiu, S., Xu, X., Cui, X., Hua, X., 2024. Change detection in remote-sensing images using pyramid pooling dynamic sparse attention network with difference enhancement. IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING 17, 7052–7067. doi:10.1109/JSTARS.2024.3374050.

Liu, M., Chai, Z., Deng, H., Liu, R., 2022. A cnn-transformer network with multiscale context aggregation for fine-grained cropland change detection. IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING 15, 4297–4306. doi:10.1109/JSTARS.2022.3177235.

Liu, S., Zhu, C., Yin, H., Qin, K., Lin, H., Huang, J., Xia, M., Weng, L., 2026. Glmamba: A global-local mamba network for efficient remote sensing change detection. IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING 19, 11344–11360. doi:10.1109/JSTARS.2026.3675679.

Lu, D., Mausel, P., Brondizio, E., Moran, E., 2004. Change detection techniques. International Journal of Remote Sensing 25, 2365–2401. doi:10.1080/0143116031000139863.

Lv, Z.Y., Huang, H.T., Li, X., Zhao, M.H., Benediktsson, J.A., Sun, W.W., Falco, N., 2022. Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective. Proceedings of the IEEE 110, 1976–1991. doi:10.1109/JPROC.2022.3219376.

Miao, L., Li, X., Zhou, X., Yao, L., Deng, Y., Hang, T., Zhou, Y., Yang, H., 2024. Snunet3+: A full-scale connected siamese network and a dataset for cultivated land change detection in high-resolution remote-sensing images. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 62. doi:10.1109/TGRS.2023.3344284.

Nemmour, H., Chibani, Y., 2006. Multiple support vector machines for land cover change detection: An application for mapping urban extensions. ISPRS Journal of Photogrammetry and Remote Sensing 61, 125–133. doi:10.1016/j.isprsjprs.2006.09.004.

Peng, X., Zhong, R., Li, Z., Li, Q., 2021. Optical remote sensing image change detection based on attention mechanism and image difference. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 59, 7296–7307. doi:10.1109/TGRS.2020.3033009.

Potapov, P., Turubanova, S., Hansen, M.C., Tyukavina, A., Zalles, V., Khan, A., Song, X.P., Pickens, A., Shen, Q., Cortez, J., 2022. Global maps of cropland extent and change show accelerated cropland expansion in the twenty-first century. NATURE FOOD 3, 19+. doi:10.1038/s43016-021-00429-z.

Radke, R.J., Andra, S., Al-Kofahi, O., Roysam, B., 2005. Image change detection algorithms: a systematic survey. IEEE Transactions on Image Processing 14, 294–307. doi:10.1109/TIP.2004.838698.

Ridd, M.K., Liu, J., 1998. A comparison of four algorithms for change detection in an urban environment. Remote Sensing of Environment 63, 95–100. doi:10.1016/S0034-4257(97)00112-0.

Saha, S., Bovolo, F., Bruzzone, L., 2019. Unsupervised deep change vector analysis for multiple-change detection in vhr images. IEEE Transactions on Geoscience and Remote Sensing 57, 3677–3693. doi:10.1109/TGRS.2018.2886643.

Song, L., Xia, M., Weng, L., Lin, H., Qian, M., Chen, B., 2023. Axial cross attention meets cnn: Bibranch fusion network for change detection. IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING 16, 32–43. doi:10.1109/JSTARS.2022.3224081.

Sun, W., Ji, Y., Wang, Y., Zhang, K., Peng, J., Li, X., 2025. Difference enhancement and interscale interactive fusion mamba for remote sensing image change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 63. doi:10.1109/TGRS.2025.3628639.

Venter, Z.S., Cz'ucz, B., Kumano-Ensby, A.L., Salzenstein, L., Simensen, T., Solvang, R., Støstad, M.N., 2026. Europe's land take and the loss of nature and cropland to artificial surfaces. Nature Communications URL: https://doi.org/10.1038/s41467-026-71931-w, doi:10.1038/s41467-026-71931-w.

Wang, H., Wang, N., Li, X., 2026. Farmcd: Agricultural information-guided gated network for farmland change detection from remote sensing images. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 64. doi:10.1109/TGRS.2026.3652586.

Wang, J., Song, J., Zhang, H., Zhang, Z., Ji, Y., Zhang, W., Zhang, J., Wang, X., 2025. Spmnet: A siamese pyramid mamba network for very-high-resolution remote sensing change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 63. doi:10.1109/TGRS.2025.3565801.

Wu, D., Chen, J., Bai, H., Yuan, L., Zhao, Q., Zheng, Y., 2025. Mdanet: A mamba-driven domain adaptation network for cropland change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 63. doi:10.1109/TGRS.2025.3633252.

Xiao, Y., Yuan, Q., Jiang, K., Chen, Y., Zhang, Q., Lin, C.W., 2025. Frequency-assisted mamba for remote sensing image super-resolution. IEEE TRANSACTIONS ON MULTIMEDIA 27, 1783–1796. doi:10.1109/TMM.2024.3521798.

Yang, M., Jiao, L., Liu, F., Hou, B., Yang, S., Jian, M., 2022. Dpfl-nets: Deep pyramid feature learning networks for multiscale change detection. IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS 33, 6402–6416. doi:10.1109/TNNLS.2021.3079627.

Zhang, C., Wang, F., Zhang, X., Wang, M., Chen, F., Wu, X., 2026. Cloud removal in remote sensing images using the g2r framework and ssmamba. REMOTE SENSING OF ENVIRONMENT 337. doi:10.1016/j.rse.2026.115321.

Zhang, C., Wang, L., Cheng, S., Li, Y., 2022. Swinsunet: Pure transformer network for remote sensing image change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 60. doi:10.1109/TGRS.2022.3160007.

Zhang, H., Chen, K., Liu, C., Chen, H., Zou, Z., Shi, Z., 2025. Cdmamba: Incorporating local clues into mamba for remote sensing image binary change detection. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING 63. doi:10.1109/TGRS.2025.3545012.

Zhu, T., Zhao, Z., Xia, M., Huang, J., Weng, L., Hu, K., Lin, H., Zhao, W., 2025. Fta-net: Frequency-temporal-aware network for remote sensing change detection. IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING 18, 3448–3460. doi:10.1109/JSTARS.2025.3525595.

Zhu, X.X., Tuia, D., Mou, L., Xia, G.S., Zhang, L., Xu, F., Fraundorfer, F., 2017. Deep learning in remote sensing: A comprehensive review and list of resources. IEEE Geoscience and Remote Sensing Magazine 5, 8–36. doi:10.1109/MGRS.2017.2762307.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.